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**M.TECH.**  
**(SEM-I) THEORY EXAMINATION 2022-23**  
**FOUNDATION OF COMPUTER SCIENCE**

**Time: 3 Hours****Total Marks: 70****Note: 1.** Attempt all Sections. If require any missing data; then choose suitably.

**SECTION A**

**1. Attempt all questions in brief.****2 x 7 = 14**

- (a) How does a hash table work?
- (b) Write different representations of graphs in the memory.
- (c) Explain the logical address space and physical address space diagrammatically
- (d) What are the various scheduling criteria for CPU scheduling?
- (e) What is the difference between a finite automaton and a pushdown automaton?
- (f) Describe the concept and use of a foreign key in RDBMS.
- (g) What is a transaction in DBMS?

**SECTION B**

**2. Attempt any three of the following:****7 x 3 = 21**

- (a) Construct the binary tree using the following traversals.  
In-Order Traversal: **D B H E I A F J C G**  
Post-Order Traversal: **D H I E B J F G C A**
- (b) Illustrate the page-replacement algorithms
  - (i) FIFO
  - (ii) Optimal Page Replacementuse the reference string **7, 0, 1, 1, 4, 2, 0, 3, 1, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1** for a memory with three frames.
- (c) Prove that the language  $L = \{0^n 1^n \mid n \geq 1\}$  is not regular.
- (d) Explain the ACID properties of transactions in DBMS.
- (e) What is database Recovery? Explain Shadow paging in detail.

**SECTION C**

**3. Attempt any one part of the following:****7 x 1 = 7**

- (a) Demonstrate the algorithms of Quick sort with the help of following numbers; also calculate its time complexity. **13, 15, 16, 1, 4, 21, 6, 17, 20, 10.**
- (b) How do collisions occur in a hash table, and how are they resolved? Explain in detail.

**4. Attempt any one part of the following:****7 x 1 = 7**

- (a) Explain the concept of Critical Section Problem. Give the conditions that a solution to the critical section problem must satisfy.
- (b) What is the role of the TLB (Translation Lookaside Buffer) in a paging system, and how does it improve performance?

**5. Attempt any *one* part of the following:**

**7 x 1 = 7**

- (a) Prove that the language  $L = \{a^n b^n c^n \mid n \geq 1\}$  is not a context-free language.
- (b) Consider the context-free grammar  $G = (V, \Sigma, R, S)$ , where  $V = \{S, A\}$ ,  $\Sigma = \{a, b, c\}$ ,  $R$  is the set of rules  
 $S \rightarrow Aa \mid bc$   
 $A \rightarrow bA \mid \epsilon$ 
  - (i) Give a leftmost derivation for the string "bbcca".
  - (ii) Give a rightmost derivation for the string "bbcca".
  - (iii) Is the string "bcbc" generated by  $G$ ? If so, give a derivation. If not, explain why not.

**6. Attempt any *one* part of the following:**

**7 x 1 = 7**

- (a) Construct a Turing Machine that will accept the Language consists of all palindromes of 0's and 1's?
- (b) Define ambiguous grammar. Check whether the grammar.  $S \rightarrow aAB$ ,  $A \rightarrow bC/cd$ ,  $C \rightarrow cd$ ,  $B \rightarrow c/d$  is ambiguous or not?

**7. Attempt any *one* part of the following:**

**7 x 1 = 7**

- (a) What is functional dependency? Explain its use in database design.
- (b) Explain the concept of Object-Oriented Database Management System (OODBMS). How is it different from a Relational Database Management System (RDBMS)?